Structural Drawing Guidelines

How to prepare structural CAD drawings so the Revit model can be built from them directly, without redrawing and without guesswork.

For	Structural consultants issuing DWG / DXF drawings to us
From	MOFA Studios Pvt. Ltd.
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1 Why this matters

We build the structural Revit model from your CAD drawings with our own software, instead of redrawing them by hand. It reads the plans, finds every column, beam, slab and wall, and matches each one to its mark and size wherever those are written in the drawing set: on the plan, in a schedule, in a detail or in a section.

The software **never guesses**. If an element has no mark, or its mark has no size anywhere in the set, it is not modelled with a made-up value. It comes back to you as a query. So the more complete and consistent the drawings, the fewer queries you get and the faster the model is ready.

We are not asking you to change your drafting style, layer names or CAD package. The software adapts to each consultant's way of drawing. What we need is what every good structural set already has: every element marked, and every mark sized.

> ### The golden rule
>
> Every column, beam, slab, wall and footing must carry a **mark** on the plan, and every mark must have a **size** written somewhere in the drawing set: on the plan, in a schedule, in a detail or in a section.

2 How the software reads your drawings

Knowing the order it works in explains most of the rules below.

1. **Sorts the sheets by their titles.** Each view is classified from its title, for example `FIRST FLOOR FRAMING PLAN`, `COLUMN LAYOUT`, `FOUNDATION PLAN`, `BEAM SCHEDULE` or `SECTION X-X`. The floor is read from the title and the file name.
2. **Finds the elements by their geometry.** Closed outlines on a column layer are columns. Pairs of parallel lines (or centrelines) on a beam layer are beams. Areas closed by beams and slab edges are slab panels.
3. **Reads the marks** written beside or inside each element, such as `C12`, `B47`, `S3` or `LW1`.
4. **Finds each mark's size** in every place it is written: a plan label `B12(230X600)`, a schedule row, a detail caption or a dimensioned section.
5. **Cross-checks every source.** When all the sources agree, the value is used. When they disagree, or say nothing, the element is flagged for a decision, and that usually means a query to you.

3 What each element needs

ELEMENT	ON THE PLAN	ITS SIZE, IN AT LEAST ONE PLACE	EXAMPLE
Column	Closed outline at its true size, with its mark beside or inside it	Column schedule (per floor if it changes), or a plan label with the size	`C12` + schedule row `C12 | 230 x 600`
Beam / girder	Drawn at its real width, mark written along the beam	Plan label with the size, beam schedule, or beam detail / section	`B12(230X600)` or `B12` with `230X600` under it
Slab	Mark written **inside every panel**, including repeated panels	Slab schedule, a thickness with the mark, or a general note	`S1` + `150 THK`, or `ALL SLABS ARE 125mm THK UNLESS OTHERWISE SPECIFIED`
Sunk slab	Outline or hatch of the sunk area	The depth written: a legend row or text inside the area	`THUS MARKED 150 MM SUNK SLAB` or `150MM SUNK` inside
Wall / lift core	Closed outline on a wall layer, with its mark	Thickness drawn true, or in a schedule / note	`LW1`, `SW2`, `RW1`
Footing	Outline on the foundation plan, with its mark	Footing schedule with size, depth and **founding level**	`F3 | 1800 x 1800 | 450 | -1.800`
Opening / cut-out	Closed outline with an **X** drawn across it	Not needed. The outline is the size.	Shaft, duct, lift well

No mark, no element. A slab panel with no slab mark in it gets no slab. A beam with no mark and no size anywhere is only built as a placeholder, marked for a decision. Please don't leave panels or beams unmarked because "it is obvious". The software cannot assume what is obvious to an engineer.

4 Marks

4.1 Write marks the same way throughout

- Use the usual prefixes. Recognised examples:
  - Beams: `B` `GB` `PB` `LB` `TB` `RB` `CB` `BK` `BKT`
  - Columns: `C` `FC` `SC` `COL`
  - Slabs: `S` `TS` `RS` `CS`
  - Walls: `SW` `LW` `RW` `CW`
  - Footings: `F` `IF` `CF` `PF` `RF` `FTG`

  Your own prefixes work too (for example `SL1` or `P1` for slabs). The software learns what they mean from your **schedule titles** and from **where the marks sit** on the plan, so always title each schedule (section 6) and use one prefix for one kind of element.
- The **same mark means the same size** on the same floor. If two beams differ, give them different marks (`B12`, `B12A`).
- The mark on the plan must be **spelt exactly as in the schedule**.
  - Capitals, spaces and a hyphen after the letters don't matter: `S1`, `S-1` and `S 1` are the same mark.
  - Everything else does. `S01` and `S1` are different marks, and so are `S1-2` and `S12`, or `B12` and `BM12`.

  Write them the same way on the plan and in the schedule.
- **Every mark on the plan has a schedule row or a size**, and every schedule row has its mark on the plan.

4.2 Place marks so they clearly belong to one element

- Beam marks run **along their beam**, close to it, nearer to it than to any other beam.
- Column marks sit inside or right beside the column outline.
- Slab marks sit **inside the panel**, not on a beam line or across two panels.
- **One mark per text object.** Write `B107` and `BP` as two texts, not `B107/BP`.
- Marks are live text (TEXT / MTEXT), not exploded into lines, and not stacked on top of another mark.

✓ Reads cleanly

`B12(230X600)` along the beam
`B12` with `230X600` directly under it
`S3` centred in its panel, `150 THK` under it
`C7` inside the column outline

✗ Causes queries

`B107/BP` (two marks in one text)
A beam mark 2 m away, nearer another beam
`S1` written on the beam between two panels
Marks drawn as exploded lines or inside a block

5 Sizes

- Sizes in **millimetres**, **width × depth**: `230X600`, `230 x 600`, `230×600`. Slab thickness as `150 THK` or `(150 THK)`.
- State the units once in the general notes: `ALL DIMENSIONS ARE IN MM`.
- A size can be written **after the mark** `GB1(230x450)`, **with a colon** `GB1:230x450`, or **directly under the mark**.
- **Varying depth** (deeper at one end): write both depths, `850/750`, and give a section showing which end is which.
- **Draw elements at their real size.** The drawn width of a beam and the outline of a column are measured and compared with the written size. A beam drawn at 230 but labelled 300 wide is flagged as a conflict.
- Don't put sizes only in layer names (e.g. `BEAM-230X600`). They are used only as a hint, and a layer name left over from an old revision causes conflicts.

6 Schedules

Schedules are where most sizes come from. The software finds a schedule by its **header words**, wherever it sits on the sheet, so the header matters most.

- Give each schedule a **title**: `BEAM SCHEDULE`, `COLUMN SCHEDULE`, `SLAB SCHEDULE`, `FOOTING SCHEDULE`. If the project has several wings or blocks, name it in the title: `BEAM SCHEDULE - WING C`.
- The **header row** needs a mark column and at least one size column, in plain words:
  - Mark column: `MARK` `BEAM NO.` `TYPE` `ID`
  - Size columns: `SIZE` `B` `D` `WIDTH` `DEPTH` `THK`

  Headers two or three rows deep are fine (`SIZE` over `B` and `D`).
- **One row per mark is best.** Grouped rows such as `B2, 4, 7` are read, but each mark in the group must share every value in the row.
- **One value per cell**, in line with its row. No "as above", no ditto marks, no blank cells for sizes.
- Real text, not exploded. Tables side by side are fine as long as each has its own title and header.
- **Columns that change size up the building**: give the size for each floor in the column schedule.
- **Footings**: the footing schedule must include the **founding level** (e.g. `-1.800` or `(-)1800`). Without it, footings cannot be placed.

7 Details and sections

- Title each section with its cut letters, `SECTION X-X`, and show the same letters on the plan where the cut is taken.
- Write the mark and size under each beam or column detail: `GB1 (230x450)`.
- Use **real dimension lines with their numbers** on shaped sections (edge beams, drop beams, upturned beams). Every corner needs a written dimension. A part shown without a dimension cannot be modelled.
- Give **levels** where they change: sunk slabs, stepped beams, landings, top of footings. Write levels as `+3.000 SFL` / `FFL`, and keep structural (SFL / TOS) and finished (FFL) levels clearly apart.

8 Drawing the plan

8.1 The set and each sheet

- Every view has a clear **title naming what it is and which floor**: `STR. LAYOUT AT 1ST FLOOR LVL`, `FOUNDATION PLAN`, `COLUMN LAYOUT`.
- The floor in the **title and the file name must agree**. A file named *3rd floor* whose title says *1st floor* is flagged.
- **Each floor in its own CAD file**, with that floor's sections and details, named with the wing and floor: *J1153 - 2ND FLOOR PLAN (C WING).dwg*. Several floors in one file can still be read, but the software then has to work out which plan, section and schedule belong together. Beam marks that repeat from floor to floor make that error-prone. Common schedules and typical details may stay in one shared file.
- **One framing plan per floor** per wing. If one plan serves several floors, say so: `TYPICAL 2ND TO 5TH FLOOR`.
- Millimetres, drawn **1:1**, nothing scaled or exploded.
- Every floor at the **same origin and rotation**, so the plans sit exactly on top of each other. Keep the grid on every plan.
- Mark parts that are **not in scope** (existing block, other building, N.I.C.) with a note or outline. Otherwise they are modelled too.

8.2 Layers

- Any layer names are fine. **Keep them the same from one issue to the next.**
- **One kind of thing per layer**: columns, beams, slabs, walls, sunk areas, openings, text, hatching, dimensions. Don't mix columns and walls, or beams and outlines.
- Only beams on the beam layer. **Don't draw shaft or duct boxes, lift walls or slab edges on the beam layer.** Every line there is read as part of a beam.
- Structural elements in model space, not inside blocks or xrefs.

8.3 Elements

ELEMENT	HOW TO DRAW IT
Columns	One **closed** outline per column at its true size. Floating columns on their own layer, or marked `FC`.
Beams	Either **both faces** at the real width, or **one centreline**. Use one way throughout. If faces are dashed under the slab and solid at edges, do it consistently. Curved beams as true arcs.
Angled beams	**Write the angle.** A beam at an angle on plan: draw it at its true angle, both ends on their supports, and write the angle beside the mark: `B14(230X600) 30°`. A sloping beam (rising from one end to the other): write the angle and the top level at both ends, `B9 SLOPING 20° TOB +3.000 TO +4.250`, and show it in a section.
Slabs	Every panel **closed** by beams, walls or a slab edge line, with no gaps at corners. One mark inside each panel.
Walls / cores	Closed outline on a wall layer, at the real thickness, with its mark (`LW1`). Not on the column layer, and not again on the beam layer.
Openings	Closed outline with an **X** across it, on an opening or cut-out layer.
Sunk areas	Closed outline or hatch, with the depth written in the legend or inside the area.
Duplicates	Nothing drawn twice on top of itself. Run `OVERKILL` before issue.

9 What happens if something is missing

IN THE DRAWING	WHAT HAPPENS
Mark and size agree everywhere	**Modelled** directly
Plan label says 230X600, schedule says 230X450	**Conflict**: not modelled until confirmed. Query to you
Mark with no size anywhere	**Missing**: placeholder at most, marked for a decision. Query to you
Slab panel with no mark	**No slab** in that panel. Query to you
Footing with no founding level	**Footing not placed**. Query to you
Mark spelt differently on plan and schedule (`S01` / `S1`)	**Not matched**: flagged as a possible match and confirmed by hand. Query to you
An unusual prefix in an untitled schedule	Its meaning may be unclear: flagged for a decision. Please title the schedule
Box drawn on the beam layer with no X	May be read as beams. Please draw openings with an X on their own layer

10 Checklist before each issue

- [ ] Every **column** has a mark, and its size is in the column schedule (per floor) or written beside it
- [ ] Every **beam** has a mark along it, and its size is in a label, the beam schedule or a detail

- [ ] Every **slab panel** has a mark inside it, and every slab mark has a thickness (schedule, label or general note)
- [ ] Every **wall / lift core** has a mark and a thickness
- [ ] Every **footing** has a mark, a size, a depth and a founding level
- [ ] Every **sunk area** has its depth written, and every **opening** is a closed outline with an X
- [ ] Marks on the plan are spelt exactly as in the schedules (same zeros, same separators), one mark per text, close to their element
- [ ] One prefix means one kind of element throughout the set (not `SB` for a slab band on one sheet and a beam on another)
- [ ] Every mark on the plan has a schedule row, and every schedule row has a mark on the plan
- [ ] Each schedule has a title and a header row (mark column + size columns), one value per cell
- [ ] Varying-depth beams show both depths and a section
- [ ] Angled beams have their angle written beside the mark. Sloping beams also have the top level at both ends and a section
- [ ] Each floor is its own CAD file, named with the wing and floor
- [ ] Sheet titles name the plan type and floor, and match the file name
- [ ] Units are millimetres, drawn 1:1, every floor at the same origin and rotation
- [ ] One kind of thing per layer, layer names unchanged from the last issue, nothing else on the beam layer
- [ ] Columns, walls, openings and slab panels are closed. No duplicates (`OVERKILL` run)
- [ ] Levels given where they change (sunk slabs, steps, landings, footings). Changes since the last issue listed

Questions about these guidelines, or would you like us to test one floor before you adopt them? Contact **MOFA Studios Pvt. Ltd.** We are happy to run a trial file through the software and send you its queries, so any gaps show up on one floor rather than on the whole project.
